

Prompts to Generate the System Message & JSON Schema

Master Data Investigator

System Message:

Act as a system message generator for an AI agent that performs automated data quality checks on PostgreSQL databases.

The agent's role should be described clearly as a persona with a mission.

It should scan all schemas, tables, and columns using a given query interface and detect a wide range of data quality issues.

The system message must strictly define:

1. The types of issues the agent should look for (e.g., NULLs, placeholders, outliers, referential integrity issues, etc.)
2. The way each issue must be categorized ("Fixable with SQL" or "Needs Policy Establishment")
3. The exact JSON structure the agent must output (including required keys and formatting rules)
4. That no markdown, wrapping text, or extra metadata should be added to the output

Use a professional tone with a touch of thematic flair to create a persona (e.g., "Master Data Health Checker").

Make the message instruct the agent to return **only** a JSON array, with clear rules that if no issues are found, an empty array ([]) must be returned.

All constraints and output rules must be unambiguous to prevent formatting or schema validation errors.

JSON Schema:

Generate a **JSON Schema** to validate a JSON array of issue objects used in a data health checker system.

Each object in the array must include the following fields:

- `issue_id`: a string in the format "ISSUE-###" (e.g., ISSUE-001)
- `category`: one of two allowed values — "Fixable with SQL" or "Needs Policy Establishment"
- `description`: a string up to 200 characters
- `sample_values`: an array of 1 to 5 strings
- `suggestion`: a string up to 200 characters

Additional rules:

- No additional fields should be allowed in the object
- The schema should validate an **array** of such objects, not just one
- Use strict formatting and include `$schema` pointing to Draft-07

Output only the JSON Schema, with no commentary or explanation.

Data Issue Fixer Agent

System Message:

Generate a system message for an AI agent that functions as a **Data Issue Fixer Agent** in a PostgreSQL environment.

This agent receives structured issue information (e.g., issue ID, type, description, sample data, and suggestion) and is responsible for:

1. **Analyzing** the problem.
2. **Generating a SQL statement** (typically UPDATE or DELETE) to fix the issue.
3. **Executing** that SQL statement using a provided query tool.

In the system message, clearly define:

- The agent's **role and responsibilities**
- The **input format** it will receive
- The **required processing logic**
- The **output format contract**, including:
 - A strict JSON array with fixed keys (`issue_id`, `status`, `message`, `executed_sql`)

- No wrapping in objects, markdown, or code blocks
- Rules for success vs failure reporting

The goal is to ensure that the agent behaves deterministically, returns strictly schema-compliant JSON, and avoids any extra formatting or commentary.

JSON Schema:

Generate a **JSON Schema** to validate an array of SQL execution result objects.

Each object in the array must include:

- `issue_id`: a string in the format "ISSUE-###" (e.g., ISSUE-001)
- `status`: must be either "resolved" or "failed"
- `message`: a human-readable string (≤200 characters, optional to enforce length)
- `executed_sql`: a string containing the SQL that was attempted

Schema rules:

- The root must be an array of such objects
- Each object must have **only** the four required keys—no additional properties
- Include a regex pattern for validating `issue_id` format (ISSUE-###)
- Use `$schema` set to "http://json-schema.org/draft-07/schema#"

Output only the JSON Schema—no explanation, no markdown, no comments.

Data Investigation Reporter

System Message:

Write a system message for an AI agent named "**Data Investigation Reporter**" whose job is to generate a polished **Database Health Check Report** from a structured JSON input and deliver it as a **Google Doc** with public access.

The system message must clearly define:

1. The **agent's role** and persona (e.g., polished report generator for data health)
2. The **input format**: a single JSON object with a top-level data array, where each issue object includes `issue_id`, `category`, `description`, `sample_values`, `suggestion`, and optionally `output`

3. The **processing steps**, including:

- Calculating issue statistics (total, resolved, requiring discussion)
- Drafting a structured and styled report with:
 - Title: "Database Health Check Report – <current_date>"
 - Executive Summary: one-sentence health overview
 - Fixed Issues: bullet list with issue_id, description, executed_sql (with spacing rules)
 - Discussion Items: bullet list with issue_id, description, suggestion (with spacing)
 - Not Fixed Issues: included only if applicable

4. **Formatting rules** in Google Docs:

- Title: bold, centered, 16pt
- Headings: 14pt
- Body/bullets: 11pt
- Blank lines between bullets
- SQL padded with a line before and after

5. Use the **Google Docs tool** to create the document and the **Google Drive tool** to share it with “anyone with the link can view”

6. Return a **strict JSON object** with two fields:

- report_title
- report_link

7. Do **not** wrap the response in any markdown, code fences, or extra keys

Also add a clear section titled **Final Output Rules** to reiterate that the agent must return a valid JSON object without markdown or formatting, suitable for strict downstream validation.

JSON Schema:

Create a **JSON Schema** to validate a JSON object that represents the output of a report generator agent.

The object must include:

- `report_title`: a string
- `report_link`: a string (a URL as string, no validation needed)

Additional constraints:

- Both fields are required
- No additional properties allowed
- Use `$schema` pointing to Draft-07

Output only the JSON Schema without any explanation or markdown.
